SCHOOL OF DIGITAL SCIENCE
UNIVERSITI BRUNEI DARUSSALAM

RULES AND REGULATIONS

Bachelor of Digital Science

For August 2022 Intake Onwards

Approved by Senate on 30 April 2022

1.0 INTRODUCTION

1.1 Application of Regulations

1.1.1 These Regulations shall apply to all students registered for the Bachelor of Digital Science degree programme offered by Universiti Brunei Darussalam (hereinafter referred to as "UBD") from the August 2020 and later intakes.

1.1.2 These Regulations should be read in conjunction with the following Regulations:

i. The existing GenNEXT Rules and Regulations for GenNEXT Undergraduates Programme; and

ii. The existing UBD Examinations Rules and Regulations for GenNEXT Undergraduates Programme.

1.2 Amendments to Regulations

These Regulations shall be subject to amendments made by the Senate from time to time. Any decision of the Senate shall be final.

2.0 ADMISSION REQUIREMENTS

2.1 Entry Requirements

To enrol in this Programme, applicants must satisfy the following entry requirements:

A minimum of **180** 'A' Level **points** for two 'A' Level OR **220** 'A' Level **points** for three 'A' Level passes in relevant science subjects, AND with at least Grade C in Mathematics at GCE 'O' Level;

OR

A minimum of **160** 'A' Level **points** for two 'A' Level OR **200** 'A' Level **points** for three 'A' Level passes in relevant science subjects, AND with at least Grade C in Mathematics at GCE 'O' Level, **AND pass an entrance examination**;

OR

An International Baccalaureate with minimum score of **26 points** including a minimum of 5 points each in at least two relevant science subjects taken at higher level, one of which must be Mathematics;

OR

An International Baccalaureate with minimum score of **24 points** including a minimum of 5 points each in at least two relevant science subjects taken at higher level, one of which must be Mathematics, **AND pass an entrance examination**;

OR

A minimum of **two** (2) **distinctions** and **four** (4) **merits** of core modules in Higher National Diploma (HND) in computer science related field, excluding Communication skills, AND with at least Grade C in Mathematics at GCE 'O' Level;

OR

A minimum of **five** (5) **merits** of core modules in Higher National Diploma (HND) in computer science related field excluding Communication Skills with at least Grade C in Mathematics at GCE 'O' Level, **AND pass an entrance examination**;

OR

At least **five** (5) **years working experience** in computer science related field; AND at least one (1) 'A' Level in a relevant subject, OR five (5) merits and /or distinctions in Higher National Diploma (HND), OR equivalent, **AND pass an entrance examination**.

2.2 English Language Requirements

At least a **Grade C6** in English Language at GCE 'O' Level or a Grade B in IGCSE English (as a Second Language) or an IELTS score of 6.0 or a TOEFL minimum overall score of 550 or a pass in the UBD Bridging English Course or Band B2 in the UBD English Proficiency Test.

3.0 DEGREE OFFERED BY FACULTY

3.1 Bachelor of Digital Science

Bachelor of Digital Science is a four-year degree programme, **offering one intake per year in August**.

3.2 Major Programmes

There are **four** (4) **major** programmes offered:

a) Computer Science
b) Data Science
c) Artificial Intelligence (AI) and Robotics
d) Cybersecurity and Forensics

The detailed programme structure for each Major is attached in **Schedule A**.

Students should seek the advice of their Academic Advisors and/or Programme Leaders prior to signing up for a module.

4.0 DEGREE STRUCTURE

No.	Type of Module	Modular Credits	Percentage	Cumulative
1	Degree Core	12	7.89	7.89
2	Major Core	48	31.58	39.47
3	Major Option	36	23.68	63.15
4	Compulsory Breadth	16	10.53	73.68
5	Discovery Year	32	21.05	94.74

6	Remaining Breadth or Option	8	5.26	100.00
	Total	**152**	**100.00**	**100.00**

5.0 GRADUATION REQUIREMENTS

Students must accumulate a minimum of **152 Modular Credits** (MCs) from a combination of degree core, major core, major option and breadth modules as specified in the following clauses and the programme structure for each major as attached in **Schedule A**.

5.1 Degree Requirements

A student must pass all **12 Modular Credits** (MCs) of **degree core** modules as prescribed in the programme structure in **Schedule A**.

5.2 Major Requirements

5.2.1 A student must pass a minimum of **84 Modular Credits** (MCs) of modules in the **major area** of study including all **48 Modular Credits** (MCs) of **major core** and a minimum **36 Modular Credits** (MCs) of **major option** modules to achieve the sufficient modular credits. A student may select the major core of another Digital Science major programme as major option to fulfil the requirements.

5.2.2 A student must pass with a maximum of **40 Modular Credits** (MCs) from **Level 1000** modules (Major and Breadth) and a minimum of **24 Modular Credits** (MCs) from **Level 4000** modules (digital science major only).

5.2.3 Detailed programme structure for each major is attached in **Schedule A**.

5.3 Breadth Requirements

5.3.1 Students must accumulate a minimum of **48 Modular Credits** (MCs) from the **breadth modules**, including 32 Modular Credits (MCs) taken in the Discovery Year.

5.3.2 Students have to take the following modules as **compulsory breadth**:

5.3.2.1 LE-1503 Communication Skills I: Academic Reading and Writing Skills (4 Modular Credits);

5.3.2.2 LE-2503 Communication Skills II: Academic Report Writing and Presentation (4 Modular Credits);

5.3.2.3 PB-1501 Melayu Islam Beraja (MIB), OR
PB-1502 Introduction to Brunei Darussalam for non-Bruneian students (4 Modular Credits); AND

5.3.2.4 MS-1501 Islamic Civilisation and The Modern World (4 Modular Credits).

5.3.3 The remaining module(s) of **8 Modular Credits** (MCs) can be taken outside the faculty or optional module within the faculty (in or outside their major programme).

5.4 Minor Requirements

5.4.1 A minor subject is optional. A student may select as his/her minor study a minimum of 24 Modular Credits (MCs) of modules from any one major subject area that differs from his/her declared major area. The minor subject can be from any other majors outside the Bachelor of Digital Science.

6.0 MINOR REQUIREMENTS FOR NON BACHELOR OF DIGITAL SCIENCE STUDENTS

6.1 Students from other degree programmes, who wish to study any one of the majors under the Bachelor of Digital Science as their minor, will need to successfully complete at least 24 Modular Credits (MCs) from the list of relevant major modules in Appendix A for their minor.

7.0 DOUBLE MAJOR REQUIREMENTS

Applicants will need to meet the following requirements to qualify in taking Double Major:

a) Must satisfy the Bachelor of Digital Science admission requirements.
b) Maintain a minimum of cGPA of 3.0 to remain pursuing a double major.
c) Declare Double Major before commencement of the fourth semester.
d) Obtain the recommendations from the Dean of both faculties to pursue a Double Major.

For other information on Double Major rules and regulations, please refer to the University GenNEXT Regulations.

7.1 Second Major outside the Bachelor of Digital Science (for Bachelor of Digital Science students)

7.1.1 Bachelor of Digital Science students who wish to take a second major outside of Bachelor of Digital Science, must successfully complete the graduation requirements of the Bachelor of Digital Science, and satisfy the second major requirements from the second major programme.

7.1.2 The requirements related to Degree Core, Major Core, Major Option, Compulsory Breadth, Discovery Year and Remaining Breadth or Option of the first major remain unchanged.

7.2 Second Major in Bachelor of Digital Science (for non Bachelor of Digital Science students)

7.2.1 Students from other degree programmes who wish to take a second major in Bachelor of Digital Science will need to successfully complete all degree and major core modules, and at least 8 MC of major option modules of the respective major programme under Bachelor of Digital Science.

8.0 DOUBLE DEGREE REQUIREMENTS

Applicants will need to meet the following requirements to qualify in taking Double Degree programme:

a) Must satisfy the Bachelor of Digital Science admission requirements.
b) Maintain a minimum of cGPA of 3.5 in both degree programmes to remain pursuing Double Degree.
c) Enrol Double Degree before end of the second semester of candidature.
d) Obtain permission from the Deans of the faculties of both degree programmes.

For other information on Double Degree rules and regulations, please refer to the University GenNEXT Regulations.

8.1 Second Degree outside the Bachelor of Digital Science (for Bachelor of Digital Science students)

8.1.1 Bachelor of Digital Science students who wish to take a Double Degree will need to satisfy the graduation requirements of the Bachelor of Digital Science.

8.1.2 Bachelor of Digital Science students are also required to satisfy the double degree requirements from the other degree programme.

8.2 Second Degree in Bachelor of Digital Science (for non Bachelor of Digital Science students)

8.2.1 Students from other degree programmes who wish to take a Double Degree in Bachelor of Digital Science as the Second Degree are required to declare a major under the Bachelor of Digital Science and satisfy the Degree Requirements and the Major Requirements of the Bachelor of Digital Science.

9.0 SIGNING UP FOR MODULES

Students must formally register for modules in order to take the examination.

Please refer to the University GenNEXT Regulations and University Examination Regulations for rules and conditions for modules registration and de-registration.

10.0 ADVANCE PLACEMENT CREDITS, EXEMPTIONS AND WAIVERS

Students who have transferred from other universities may qualify for exemptions with the Faculty's approval. Credit Transfer Regulations (Regulations and Procedures for Credit Transfer for GenNEXT Bachelor Degree Programmes) will be followed.

11.0 DISCOVERY YEAR PROGRAMME

11.1 Students will undertake a Discovery Year comprised of spending **two periods of time equivalent to two semester**s in any two of the activities: study abroad programme (SAP), internship, community outreach programme, or innovation or incubation projects as specified in the GenNEXT Rules and Regulations for GenNEXT Undergraduates Programme. Students are only allowed to do the activities that support the programme objectives and learning outcomes subject to approval by the Programme Advisors.

11.2 The schedule of each Discovery Year period **may be during a teaching semester, or during the semester break**. A student shall work out the actual activities and schedules with the Programme Advisors subject to the availability of places as well as the plan of the degree programme.

11.3 For Study Abroad Program (SAP) module, the modular credits count towards the credit requirement, but they will not be counted towards CGPA.

11.4 Internship, community outreach programme, innovation/ incubation projects and summer/Co-op training are graded as a satisfactory/unsatisfactory basis. Students who do not satisfy internship, community outreach programme or innovation/ incubation projects will have to fulfil their modular credit requirements for graduation by reading outside Faculty breadth modules in UBD.

11.5 Students may, in special cases, and on medical and compassionate grounds, be exempted from the Discovery Year programme. Approval must be sought from the Dean for such exemptions and shall be considered on a case by case basis. For such students, they will take outside Faculty breadth modules offered by UBD.

For other rules and regulations on Discovery Year matters, please refer to the Faculty Discovery Year Coordinator or Programme Discovery Year Advisor.

12.0 ACADEMIC ADVISORS

All academic staff involved in a programme will act as Academic Advisors to the students in the respective programme, where his/her roles and responsibilities include:

12.1 Acting as a mentor to the assigned group of students;

12.2 Counselling and discussing with students on any issues related to academic progress whenever required;

12.3 Guiding students in selection of option and breadth modules for completion of degree with broad knowledge of the relevant academic areas and disciplines

12.4 Keeping track of students' progress throughout the whole duration of the Degree; and

12.5 Liaising with Dean of Students where and when necessary.

13.0 ASSESSMENT

13.1 Students reading a module shall be assessed on the overall performance in the module through a combination of continuous assessment and/or written examinations and/or other suitable methods.

13.2 Some modules may have 100% continuous assessment. The continuous assessment grade shall be a combination of grades from different pieces of class works, laboratory works, and/or field works. The specific methods and proportions of assessment components shall be stated clearly on the published module content/description of each module.

13.3 Please refer to the University GenNEXT Regulations and University Examination Regulations and Procedures regarding transfers and failures.

14.0 PLAGIARISM

14.1 Students should be aware that plagiarism is a form of academic misconduct that puts at risk their continuation as a student in UBD. The concept that is central to the allegation of plagiarism is that the student intended to plagiarise the work or ideas of another person.

14.2 Students will have to follow the procedures set out in the UBD Examinations Rules and Regulations for GenNEXT Undergraduates Programme and or any other Policy governing plagiarism in UBD.

15.0 EXAMINATION

Regulation governing examination on the Degree Programme should be read in conjunction with the UBD Examinations Rules and Regulations for GenNEXT Undergraduates Programme.

16.0 SUPPLEMENTARY EXAMINATION

There shall be no supplementary examination.

17.0 BARRING OF STUDENTS

17.1 Students are strongly encouraged to attend all classes. Attendance in lectures, tutorials and laboratory classes is compulsory. If a student fails to attend 60% of the tutorial/laboratory classes for a particular module, he/she shall be barred from taking the examinations for that particular module.

17.2 The Dean may, on the recommendation of the Programme Leader, give recommendation to the examination office to bar students from taking the prescribed examination for any module on grounds of unsatisfactory attendance, unsatisfactory progress in coursework and misconduct in laboratory classes or fieldwork.

17.3 Please refer to the University Examination Regulations and Procedures for other rules and regulations regarding barring of students.

18.0 TERMINATION

18.1 The Regulations of transfer between Programmes and Termination of Candidature, stated in the UBD Examinations Rules and Regulations for GenNEXT Undergraduates Programme shall apply to students of this Programme.

18.2 Students who are terminated from the Programme are not allowed to reapply in the same Programme.

19.0 CLASSIFICATION OF DEGREE

Award of the Degree will be conducted in accordance to the GenNEXT Rules and Regulations for GenNEXT Undergraduates Programme.

APPENDIX A: List of modules for minor in Bachelor of Digital Science

Minor in Cybersecurity and Forensics can be awarded to students that have obtained 24 Modular Credits (MCs) from any modules listed below:

No.	Code	Module	MCs
1.	ZZ-1102	Programming Fundamentals	4
2.	ZC-1201	Computer Architecture and Organisation	4
3.	ZS-1201	Introduction to Cybersecurity	4
4.	ZS-2201	Digital Forensics	4
5.	ZS-2202	Computer Security	4
6.	ZS-3201	Cybersecurity System Lab	4
7.	ZS-3301	Block Chain Technology	4
8.	ZS-4302	Cyber Criminology	4
9.	ZS-4303	Cybercrime and Investigation	4
10.	ZS-4305	Application and System Software Security	4
11.	ZS-4307	Digital Supply Change	4
12.	ZS-4309	Emerging Technologies in Cybersecurity	4

Minor in Computer Science can be awarded to students that have obtained 24 Modular Credits (MCs) from any modules listed below:

No.	Code	Module	MCs
1.	ZZ-1102	Programming Fundamentals	4
2.	ZC-1201	Computer Architecture and Organisation	4
3.	ZC-2201	Database Design	4
4.	ZC-2203	Computer Networks	4
5.	ZC-2205	Data Structures and Algorithms	4
6.	ZC-2204	Software Engineering	4
7.	ZC-2202	Operating Systems	4
8.	ZC-3302	Internet Programming and Development	4
9.	ZC-3313	Game Programming	4
10.	ZC-4302	Internet Application Development	4
11.	ZC-4303	Large Scale Software Development	4

Minor in Data Science (DS) can be awarded to students that have obtained 24 Modular Credits (MCs) from any modules listed below:

No.	Code	Module	MCs
1.	ZZ-1102	Programming Fundamentals	4
2.	ZD-1201	Introduction to Data Analytics	4
3.	ZA-3202	Machine Learning	4
4.	ZD-3201	Data Analytics Lab	4
5.	ZD-4201	Big Data Analytics and Data Warehouse	4
6.	ZD-2301	Data Engineering	4
7.	ZD-4301	Bioinformatics	4
8.	ZD-4302	Business Technology Management	4
9.	ZD-4303	Digital Business Intelligence	4
10.	ZD-4309	Emerging Technologies in Data Analytics	4

Minor in Artificial Intelligence & Robotics (AIR) can be awarded to students that have obtained 24 Modular Credits (MCs) from any modules listed below:

No.	Code	Module	MCs
2.	ZZ-1102	Programming Fundamentals	4
3.	ZA-2201	Artificial Intelligence	4
4.	ZA-2202	Physical Computing for Intelligent Systems	4
5.	ZA-2203	Robotics Systems	4
6.	ZA-2204	Electronics for Intelligent Systems	4
7.	ZA-3201	AI Application Lab	4
8.	ZA-3202	Machine Learning	4
9.	ZA-4301	Autonomous Mobile Robots	4
10.	ZA-4302	Deep Learning	4
11.	ZA-4303	Graph Models	4
12.	ZA-4304	Multi Agent Systems	4
13.	ZA-4305	Information Retrieval	4
14.	ZA-4306	Natural Language Processing	4
15.	ZA-4307	Artificial Life and Evolutionary Computation	4
16.	ZA-4308	Machine Perception	4
17.	ZA-4309	Emerging Technologies in Intelligent Systems	4
18.	ZA-4310	Smart Systems for IR 4.0	4
19.	ZA-4311	Precision Agriculture	4